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Survey paper

A comprehensive overview of topic modeling: Techniques, applications and challenges

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ABSTRACT

The growing need of gaining valuable information from large collections of text data has attracted a considerable interest from researchers in the field of natural language processing (NLP) and machine learning. Topic modeling is an unsupervised NLP technique that aims to extract hidden themes within a corpus of textual documents. This paper provides a thorough and comprehensive review of topic modeling techniques from classical methods such as latent sematic analysis to most cutting-edge neural approaches and transformer-based methods. We also introduce topic modeling applications in various domains such as text classification, bioinformatics and sentiment analysis. Real-world case studies are examined to show the versatility of topic modeling in discovering hidden structures within text and unstructured data to facilitate the process of knowledge discovery. The authors also address current challenges facing topic modeling such as interpretability and scalability, and discuss future research directions. This review aims to provide both academic researchers and practitioners with a good understanding of topic modeling techniques, applications, datasets, and research trends.

1. Introduction

The large amounts of user generated data on the internet make manual processing of texts to extract valuable information very difficult in terms of time and cost. NLP techniques play an important role in the process of gaining valuable knowledge from text data. Topic modeling is one of the most extensively used NLP tools for processing large collections of unstructured textual documents. Topic models are widely applied in various domains, including text classification, sentiment analysis, and information retrieval, etc. Researchers in such domains have developed many techniques to automate topic extraction and modeling from large corpora since the introduction of latent semantic indexing (LSI) in 1990 [1]. Topic models are designed to discover hidden topics in a collection of documents following two main assumptions: (i) each document in the collection is represented as a set of various topics and (ii) each topic is considered as a distribution over words. In other words, A document is a distribution of different topics with proportions, and these topics, in turn, are a distribution of words with probabilities.

Text representation learning is another fundamental NLP technique that enables machines to effectively process and understand textual data [2,3]. By learning document representations, topic modeling as one of these techniques aims to uncover latent themes in large corpora. This allows topic modeling to be applied for different tasks such as document categorization and text clustering, which makes it a powerful tool for organizing and interpreting large volumes of textual data. However, this approach is just one of many approaches used for text representation learning. Several language models such as word2vec [4], BERT [5] and GPT [6] play a key role in representing text. These language models usually learn contextual embedding for individual words, while topic modeling focuses on thematic structures to emphasize document-level patterns [7]. This makes it very useful for high-level document understanding. On the other side, and generally speaking, language models are effective for fine-grained word-level tasks like sentiment analysis [8,9].

Large Language Models (LLMs) like BERT and GPT have introduced new possibilities in topic modeling by capturing contextual word embeddings and semantic nuances often ignored by traditional models [7,10]. Nowadays, Prompt-based Topic Modeling [11–13] has emerged as an advanced method that exploits the capabilities of LLMs to generate topics. Unlike traditional methods, the new approach leverages the strength of LLMs by prompting them to directly produce topic

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distributions [13]. Moreover, recent advancements in the field have incorporated transformer-based architectures for dynamic topic extraction, known as Transformer-based Topic Modeling (TTM) [14,15]. This approach adapts topics either over time or cross different contexts within a corpus, which is very important for Dynamic Topic Modeling [16,17].

Despite the differences between both approaches, they can be seen as complementary. For instance, due to the contextual embedding, language models face difficulties in producing document-level thematic structures directly. This is not a problem for topic modeling since it excels at summarizing large text corpora through broader thematic insights. Thus, it can serve as a preliminary framework for deeper context-based analysis with language models. Therefore, the combination of both approaches can lead to more effective understanding of text data, particularly for applications that require document-level themes and detailed word-level context. Language models have proven their significance in generating semantically rich topics. However, these models are computationally expensive and can sometimes generate overly broad topics.

Several surveys and review papers on topic modeling have been presented in recent years. Alghamdi and Alfaqli [18] have introduced a review of topic modeling techniques in text mining and presented a chronological evolution of topic models over time. In [19], Sharma et al. surveyed topic models based on a classification of being supervised or unsupervised, LDA or non LDA variant, and finally bag-of-words-based (BOW) or sequence-of-words-based. In[20], the authors presented a full overview of topic modeling techniques and topic evolution models in text classification. Jelodar et al. [21] have presented a survey paper on LDA and its applications across different fields. Another review paper was conducted by Qiang et al.[22] to summarize topic modeling methods for short text documents. The authors categorized these techniques into three main classes: Dirichlet multinomial mixture, global word co-occurrences, and self-aggregation-based models. They compared these models on different benchmarking real world datasets with the results of long text topic modeling. In [23], the authors provided an evolutionary survey on topic modeling from 1990 until 2022. They performed an experimental comparison on different topic models and algorithms in terms of evaluation metrics such as coherence and diversity. In [24], the authors classified topic models into four categories: algebraic, probabilistic, neural, and fuzzy models.

Topic models are widely applied in many fields such as social media analysis[25], information retrieval [26], bioinformatics [27], business intelligence [28], etc., to extract valuable information from textual data. Topic modeling techniques have gained more interest in recent years due to their importance in many areas that are related to text mining and natural language processing. Researchers have used topic models to identify the main issues and themes from historical newspapers [29]. Sleeman et al. [30] have implemented topic modeling techniques to analyze trends in climate change reports over 30 years of time period. Topic models are also used effectively in information retrieval (IR) to enhance query expansion and smoothing documents for search engines [31].

To the extent of our knowledge, the majority of existing reviews have not provided a comprehensive and global overview of topic modeling while others have only focused on surveying the evolution of topic modeling over time. Some surveys have been limited to specific applications, such as text mining. In contrast, our review takes a wider perspective, encompassing current approaches, techniques, datasets, challenges, and applications of topic modeling across various domains.

This paper is designed to survey the recent advancements in the field of topic modeling, review its most cutting-edge techniques, and shed light on its applications across different domains. The main objective of this survey is to summarize research works that have been published in the last three decades for both academics and developers searching to gain a comprehensive overview of this field. In this survey, the authors define topic modeling terminology and its general process beginning with data collection and preprocessing to topic extraction and modeling. In this overview, we also delve into the foundational concepts of topic modeling techniques. The authors have classified these models into three main categories: algebraic, probabilistic, and neural topic models. On a second level, we classified these techniques based on the type of vectorization: bag of words or word embedding models. Furthermore, this survey examines the various applications of topic modeling in several fields such as information retrieval, sentiment analysis, document classification, etc. The authors have also discussed some challenges that topic modeling is still facing nowadays.

The rest of the paper is organized as follows: Section 2 presents some benchmarking datasets. Section 3 introduces vectorization models used in text representation. In Section 4, we review topic modeling techniques classified in three main types: algebraic, probabilistic, and neural. In Section 5, we present different evaluation metrics to assess the performance of topic models. Section 6 presents experiment results of comparison of some topic models. In Section 7, we cite some applications of topic modeling techniques. In Section 8, we unveil challenges facing topic modeling in text mining. Finally, conclusion and future research directions are discussed in Section 9.

2. Benchmark datasets

Data collection is considered to be the most crucial step in any topic modeling task. It is always valuable and desirable to gather textual data from different sources in different formats, because the following NLP tasks in the process of topic modeling highly depend on the quality of the collected data as well as the length of texts and the size of the dataset. Many publicly available datasets were collected and preprocessed for the purposes of research, but they do not always address the typical problem of a certain topic modeling task. For instance, 20Newsgroups is a textual dataset that contains 20000 documents covering various topics from 20 news groups [32]. BBC News is another frequently used dataset for topic modeling applications such as text summarization [33]. The dataset contains 2225 articles which are classified into five different labels: business, entertainment, politics, sport, and tech. Reuters-21578 is another corpus of news articles that contains more than 10360 documents and a vocabulary of 29,930 words [34]. This corpus is widely used by academic researchers as a benchmarking dataset for training and testing topic modeling techniques to perform NLP tasks such as document clustering, text summarization, and sentiment analysis. Table 1 shows examples of used datasets in various topic modeling tasks and Fig. 1 illustrates the general process of topic modeling.

For Text classification tasks, both supervised and unsupervised methods can be used. While supervised classification relies on labeled data to categorize text, unsupervised methods, such as clustering, group similar texts without predefined labels. Thus, clustering is a valuable solution when exploring natural groupings within unstructured data or in cases where labeled data is scarce. It allows the discovery of latent structures and topics within data, which makes it useful in exploratory analysis and knowledge discovery [15,35]. However, since text classification is typically a supervised task, most datasets are labeled. In addition to using the unlabeled version of these datasets, researchers can collect more unlabeled data for unsupervised tasks such as clustering. The advantage of using labeled datasets for clustering is the once this task is performed, researchers can compare the clustering results against the performance of supervised algorithms to assess how well the clustering captures the underlying structure of data [36]. Researchers can use also some unlabeled datasets such as BooksCorpus [37], Text8 [38], and OpenWebText [39] either for text classification, topic modeling or both. The combination of topic modeling and clustering provides a range of applications. For instance, the ability to identify themes with clustering's capacity to group similar items. This can help Achieving deeper insights in fields such as document categorization and automatic term extraction [36,40,41]. The last three studies in Table 1 have combined topic modeling with unsupervised techniques for different tasks.

Table 1
Topic modeling tasks over different datasets.

References	Dataset	Task	Topic Model	Results
[42]	IMDB Movie Reviews	Sentiment Analysis	LDA	Good topic partitioning and high accuracy on sentiment analysis results on both document and word level
[43]	ACTER	Automatic term extraction	NMF	NMF outperformed 4 out of 6 baseline models and NMF-based algorithms depend on the corpus imbalance
[44]	Twitter	Text classification	BERTopic	Transformer based BERTopic model outperformed other baseline models based on many evaluation metrics such as F1, precision, recall, and accuracy.
[45]	Reuters−21578 and 20Newsgroups	Document categorization	Doc2vec	Good F measures and time computation and their rule-based approach outperformed other algorithms such as JRip, OneR, and ZeroR.
[46]	Scientific papers from IEEE, Scopus, and Springer databases	Opinion mining	Word2vec+LSA	High accuracy and diversity score of extracted topics in terms of qualitative analysis The proposed approach outperformed other existing methods on both quantitative and qualitative levels
[47]	20Newsgroups	Text summarization	CSTM	The authors have demonstrated through experimental results that CSTM has better performance than a two-stage approach based on LDA.
[48]	COVID−19 Hespress comments[a]	Topic-based sentiment analysis	LDA+CAMeL	The proposed model uses LDA to extract topics from comments and CAMeL to classify sentiment polarity. Results showed that negative sentiments regarding all extracted topics were highly negative.
[40]	Collected Dataset	Document Categorization	BERTopic + LDA	The proposed model used two topic modeling techniques, BERTopic and Latent Dirichlet Allocation (LDA). For BERTopic to extract meaningful topic, it leverages Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN).
[41]	Collected Dataset	Automatic Term Extraction	LDA + Concept hierarchy learning	This study combined topic modeling, statistical and linguistic processing, and probabilistic hierarchy learning to identify and structure concepts from a large unstructured text corpus, all within an unsupervised learning framework
[36]	Collected Dataset	Document Categorization	Doc2vec + Spherical clustering	The proposed methodology uses Doc2Vec for vectorization, followed by spectral clustering for unsupervised grouping of articles.

[a] https://github.com/HankarM88/Hespress_COVID-19_Dataset

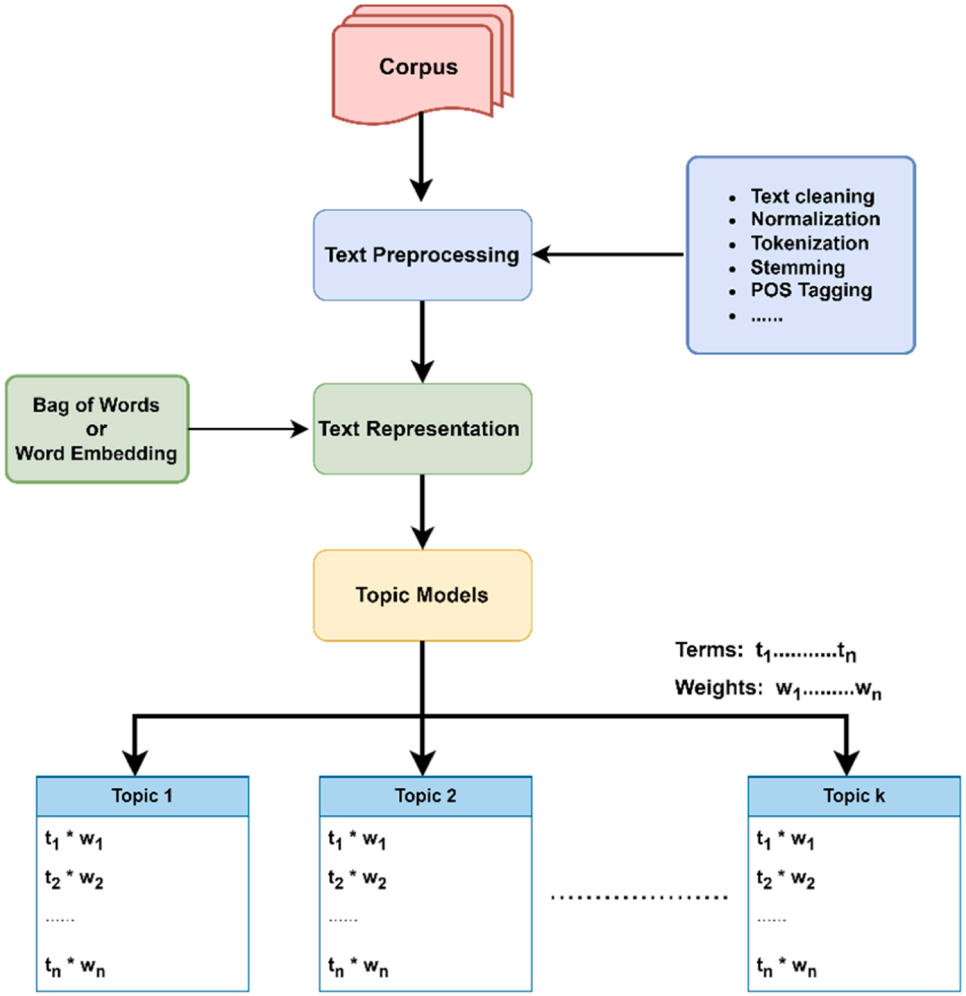


Fig. 1. General process of topic modeling.

3. Text representation

This section provides an overview of the most commonly used text representation techniques for topic modeling and highlights the differences between applying topic modeling to long and short texts. Additionally, it explains why short texts present unique challenges for NLP tasks, such as topic modeling. To the best of our knowledge, other topic modeling surveys do not include this discussion. Topic modeling techniques vary significantly in effectiveness depending on the length and richness of the input text. According to Birjali et al. [49], short texts such as tweets, customer reviews, and abstracts present unique challenges in NLP tasks. In the context of topic modeling, short texts often contain limited context which makes extracting coherent topics very difficult. This is due to the sparsity of the vocabulary, causing many words not to appear frequently which enables the identification of meaningful topics [50]. On the other hand, long texts like full articles and documents often have more structured narratives, more context, and a richer vocabulary, making it easier to identify distinct themes and topics. While traditional techniques like LDA and similar models can effectively capture underlying themes in non-short texts due to the presence of larger text segments, they may not perform well with short texts because of insufficient information. Several studies are proposed to overcome the data sparsity challenge [7,50–52]. Preprocessing on the other side appears as a significant problem. Usually, short texts need aggressive preprocessing [49] like combining similar terms or using synonyms to enrich the text data. While this step is important for both text types, expanding text is not a necessity for longer texts.

After the preprocessing phase, the text needs to be converted from its natural form into a machine-readable vectorized representation before performing any type of analysis, i.e., topic modeling. The efficiency of these representations directly affects the performance of further NLP tasks [53]. This section introduces the most two common text representations used to build topic models, highlighting their differences as well as the advantages and disadvantages of each.

3.1. Bag of words (BOW)

Bow model is the basic way of representing texts into numbers to build topic models and machine learning models. BOW representation involves the construction of a vocabulary of unique words from the collection of input documents, and a certain metric to measure each word in each document of the corpus. Most common used metrics to compute word weighting are term frequency (TF) and term frequency-Inverted document frequency (TF-IDF). The first determines the importance of the term to a specific document and the second undermines the frequent words (such as stop words) in the collection to assess the presence of rare words that may be important. To compute TF-IDF weighting score, we use the following formula:

$$\text{tfidf}_{w,d} = \text{tf}_{w,d} \times \log\left(\frac{\text{N}}{\text{df}_w}\right) \quad (1)$$

Where $\text{tf}_{w,d}$ is the total number of occurrences of the word w in a document d, N denotes the total number of documents in the corpus, and df_w is the number of documents containing the word w.

The main limitation of BOW model is that this representation discards the order of words in text sequences which is an important factor in capturing their context. BOW causes the loss of valuable information by disregarding the contextual aspect of words within a document. Researchers have proposed a hybrid model that combines both BOW and n-gram models to improve the latent topic semantics, and the local Markovian dependencies in the text [54]. By implementing the proposed model, they were able to achieve 10 % of reduction in perplexity and 2 % of reduction in the relative word error rate (WER) for transcribing broadcast news in English. Another drawback of BOW representation is that it does not capture semantic meaning of words and treats all terms as independent entities in the document which leads to ignore the semantic relationship between them. Moreover, BOW vectorization generates high dimensional sparse vectors when dealing with large corpora which increases the computational complexity and memory requirements. Additionally, and in many cases, BOW fails to handle synonymy treats words with the same meaning (e.g., “happy”, “joyful”) as unrelated.

3.2. Word embeddings

The limitations of bag-of-words models, even if improved using n-grams, can partially or totally fail to capture the context and the semantic aspect of words within a document. Therefore, word embeddings [55] are designed to solve this problem by semantically considering the order in a sequence of words. Word embedding models try to capture the meaning of a text and find similarities between the words that construct the vocabulary of a corpus in the vector space[56]. These models are built on top of artificial neural networks (ANN) and trained on large datasets in order to capture the contextual meaning of a word unit based on its surrounding words, and vice versa. In Fig. 2, we show a basic example of word embedding model visualized in three-dimensional space.

Each word in the embeddings space is represented as a vector of real numbers of a predefined dimension. In a space of word vectors, we can identify the most similar words of a certain word in the vocabulary using the cosine similarity method or the Euclidian distance method. These techniques help to cluster related words, classify texts and categorize documents, and find latent themes or topics within the text.

3.2.1. Word2Vec

Word2vec [4] was first developed by Google researchers in 2013 to generate distributed and continuous dense vector representations of words from a massive textual corpus. It has the ability to capture contextual meaning and semantic similarity of words within the vocabulary of unique words from a collection of documents. The size of word vectors can be specified to make the dimensionality of dense vector space much lower than the high-dimensional sparse vector space built using traditional BOW models. Word2Vec is a prediction-based model that can be applied in two architectures: continuous bag-of-words (CBOW) and Skip Gram [57]. Both CBOW and Skip Gram models are built on top of neural networks which are pretrained to learn the mappings between words in the constructed vector space.

CBOW architecture is trained to predict the center word based on its surrounding words with a fixed window size. Researchers in [4] noticed that CBOW runs much faster than skip-gram and learns syntactic mappings between words much better. Skip Gram model tries to do exactly the opposite of what CBOW model does. This means that the model

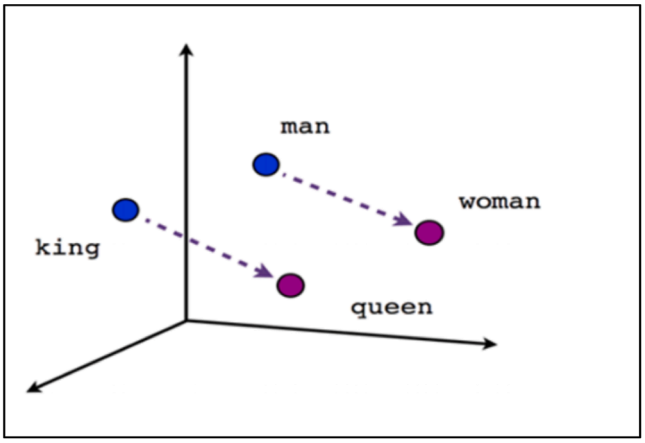


Fig. 2. Example of words embedding visualization.

predicts the surrounding words based on the center word. The key parameters need to be tuned while training a Word2Vec model. One of these parameters is the number of the embedding dimensions and the other is the size of the context window, i.e., how many words before and after the center word should be used as a context for training the word embeddings. Increasing the dimensionality of word embeddings typically requires more training data and each dimension should capture the contextual aspect of words. Fig. 3 illustrates the main difference between CBOW and Skip-gram architectures.

3.2.2. GloVe

GloVe stands for Global Vectors which is a word embedding model designed by Stanford researchers [58] to create dense word vectors. However, the way that these vectors are created is different from Word2Vec model. The model is trained to learn from a word co-occurrence matrix to create a space vector with meaningful sub-structures. The dimension of the co-occurrence matrix is given by $V \times V$ where V is the vocabulary size of the whole corpus. Each element of this matrix corresponds to the number of times the words of the vocabulary occur together within a pre-specified context window moving across the entire corpus. GloVe learns vector embeddings by minimizing the error rate between the predicted co-occurrence statistics by the model and the observed co-occurrence statistics in the training corpus. The model has a set of hyper-parameters to choose and tune like the vector embedding dimension of words and the context window size. GloVe vectors are similar to those created from Word2Vec model but the co-occurrence matrix method rather than prediction-based method [59]. GloVe model computes statistics over large pre-specified context windows better than Word2Vec and captures long-term dependencies. The performance of both GloVe and Word2Vec models depends on many factors like the type of training data and the evaluation criteria.

3.2.3. FastText

FastText model [60] was first presented by Facebook research team in 2016 as an extended and improved version of Word2Vec. The model is designed to overcome some limitations in Word2Vec and GloVe models like ignoring the morphology of words. Furthermore, FastText is based on Skip Gram model where every word is represented as a bag of n-gram characters. Each vector representation is related to each n-gram character and the words in the vector space are represented as the sum of these representations. This allows to represent the morphology and lexical similarity of words, and being able to formulate vectors for unseen words. FastText is capable to learn word representations and perform fast and accurate text classification tasks on top of these word embeddings. Athiwaratkun et al. [61] proposed a probabilistic version of FastText that integrates Gaussian mixture models (GMM) to represent each word as a Gaussian mixture of K different meanings. This representation is able to capture the structure of sub-words, different word senses, and can provide better representation of rare or unseen words.

3.2.4. Doc2vec

Doc2Vec is a paragraph-based model that extends Word2Vec to capture document-level semantic representations. It is a neural network-based model that learns continuous distributed representations of documents to enable the identification of similar documents and the extraction of topics from unstructured text data. The key idea behind Doc2Vec is to represent each document as a fixed-length vector often referred to as a document embedding or a document vector [62]. This vector is trained to capture the semantic meaning of the document by considering the context of its words. The algorithm takes into account both the words in the document and their order allowing to capture both local and global context information. The training process involves feeding the algorithm a set of labeled documents and optimizing the model to predict the context words given a specific document or predict the document label given a context window of words. This unsupervised learning process aims to learn meaningful representations of the underlying topics in a collection of documents. Doc2Vec has been widely used in various applications including text classification, document similarity analysis, and information retrieval [63]. This representation offers several advantages over other models such as the ability to handle out-of-vocabulary words and the capability to generate document embeddings that can be used for downstream tasks.

3.2.5. Other models

Many other models of text representations were proposed in recent years to overcome the limitations faced in the previous representations. For instance, Devlin et al. [64] proposed a Bidirectional Encoder Representations from Transformers (BERT) which is a multilayer neural networks transformer based on parallel attention [65]. Peters et al. [66] proposed a deep contextualized model named embedding from language models (ELMs) that captures word embedding from the entire sentence. The model implements bi-directional recurrent neural network (RNN) and is trained to create embeddings by looking at the next and previous words in the sentence. Wissam et al. [67] have proposed AraBERT, a pre-trained BERT model designed specifically for Arabic language to achieve the state-of-the-art results showed in other languages such as English to perform many NLP tasks.

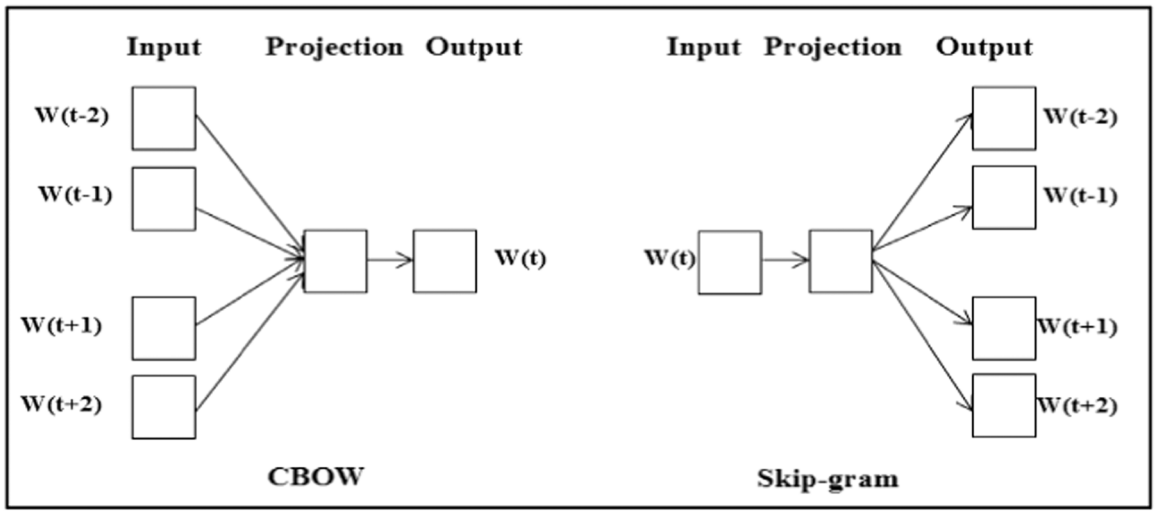


Fig. 3. CBOW and Skip-Gram architectures [25].

4. Topic modeling techniques

In natural language understanding (NLU), there is a variety of methods available for extracting valuable insights from a given text whether it is a word, a sentence, or a document. One of the most useful ways to understand a text at the document level is to analyze its main topics. The process of learning, recognizing, and extracting these topics from a collection of documents is called topic modeling. In Fig. 4, we present the overall taxonomy of topic modeling techniques.

4.1. Latent semantic analysis

Latent semantic analysis (LSA) or latent semantic indexing (LSI) is considered the most foundational technique of topic modeling. This model traces its origin back to 1990 and was introduced by Deerwester et. al [1] to index and retrieve documents from large databases. LSA functions by separating the document-term matrix (DTM) generated from the corpus into three main matrices, namely, document-topic matrix, topic matrix, topic-term matrix. Given N documents in the collection and a vocabulary of size M, then DTM is constructed in a way that each row represents a document and each column represents a word from the vocabulary. LSA takes the DTM matrix as an input and then decompose it using a matrix factorization method named singular value decomposition (SVD) to create document-topic matrix, diagonal matrix of singular values, and topic-term matrix. The first matrix generates the main topics from documents, the second matrix generates topics, and the third matrix generates the words constructing each extracted topic. Since DTM matrix is very sparce, reducing its dimensionality using SVD extracts latent topics that capture the relationships between words and documents. Given a DTM matrix called X to be decomposed using SVD. The result is an output of three different matrices: X = W*S*P where S is the diagonal matrix of singular values. The Fig. 5 below illustrates the implementation of SVD in LSA model.

X is a term-by-document matrix decomposed into three other matrices S, W, and P. m denotes the number of terms (or unique words) in term-by-document matrix X, n denotes the number of documents in the corpus, and k denotes the number of latent concepts or topics. Document vectors in the matrix W and word vectors in the matrix P are typically used to compute cosine similarity between different documents in the collection and word similarities.

4.2. Probabilistic latent semantic analysis

Probabilistic Latent Semantic Analysis (PLSA), also known as Probabilistic Latent Semantic Indexing (PLSI), is a probabilistic topic model that was introduced by Jan Puzicha and Thomas Hofmann [68] in 1999 to handle some drawbacks of LSA. The main difference between the two topic models is that PLSA uses a probabilistic method instead of SVD. In PLSA, a probabilistic method is used with latent topics to generate data from DTM. Given D a set of documents, and W a set of vocabulary words, P(D,W) is approached in a way that for any document d and word w, P(d, w) corresponds to an element of DTM matrix. Knowing that each document is a mixture of topics and each topic is a distribution of words. PLSA transforms this as:

- The probability P(z|d) that a topic z is present in a document d
- The probability P(w|z) that a word w is present in a topic z

The graphical representation of PLSA model is depicted in Fig. 6.
The joint probability of seeing a document and word together is:

$$P(D,W) = P(D)\sum_{Z} P(Z|D)P(W|Z) \tag{2}$$

This equation estimates the likelihood of seeing a given word in a document in the corpus based on the distribution of latent topics in that document. In this case, P(D), P(Z|D), and P(W|Z) are the parameters of the model. P(D) is a corpus-level parameter and P(Z|D), P(W|Z) are represented as multinomial distributions that are trained using the expectation-maximization (EM). EM is an optimization algorithm to find the parameter estimates for a model based on latent variables (i.e.; the topics). Interestingly, the probability P(D,W) is also obtained using a different set of three parameters:

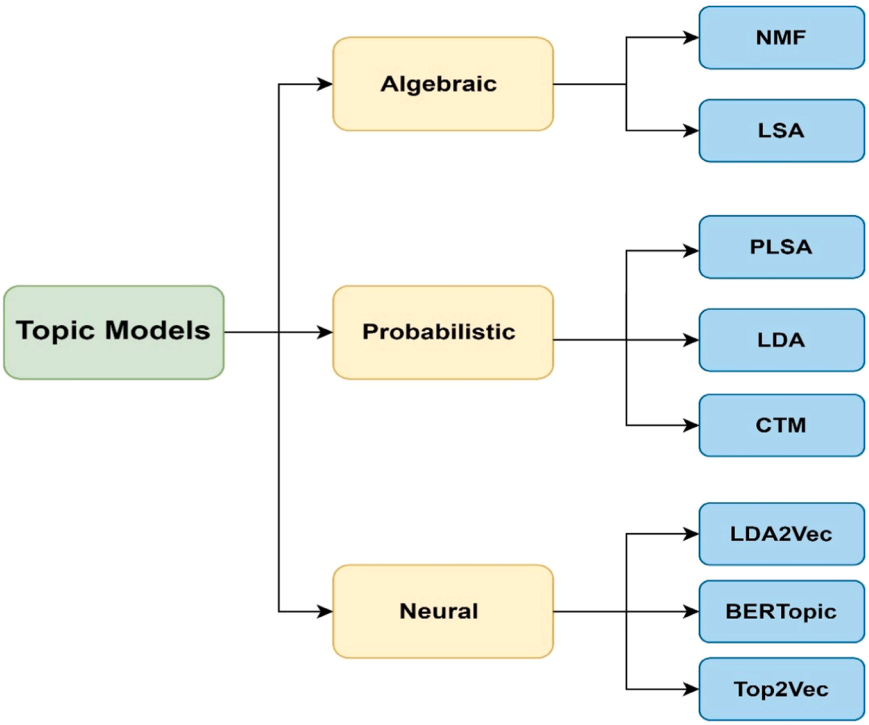


Fig. 4. Topic modeling techniques taxonomy.

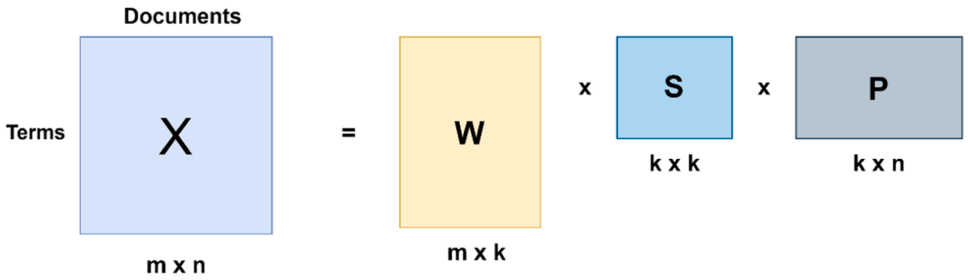


Fig. 5. Application of SVD in LSA model.

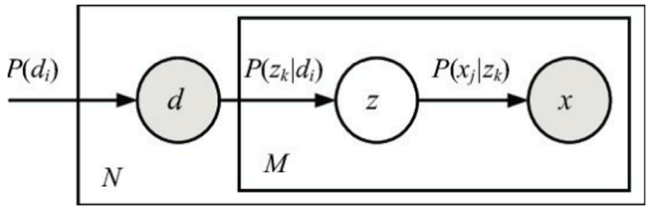


Fig. 6. Graphical representation of PLSA model.

$$P(D,W)=\sum_{Z}P(Z)P(D|Z)P(W|Z) \tag{3}$$

The PLSA model can be viewed as a generative process. In the first step, each document is parametrized with P(d), then topics are generated with P(z|d), and finally words are generated with P(w|z). PLSA is based on an aspect model which is defined as a latent variable model for co-occurrence data that associates unobserved class variables with each observation [69]. PLSA model is designed to overcome some LSA limitations by solving the problem of disambiguation and polysemy (i.e., words with multiple meanings), and by grouping similar words that share a common context [70]. PLSA is widely applied in many real-world applications such as computer vision and recommender systems. However, as the number of parameters increases linearly with the size of the corpus, PLSA may suffer from overfitting problems [71].

4.3. Latent dirichlet allocation

Latent Dirichlet allocation (LDA) was first introduced by Blei et al. [72] in 2003 as a generative Bayesian model to identify hidden topics from a collection of documents. LDA remains one of the most popular topic modeling techniques that are widely applied in real-life applications due to its promising and accurate results. In a collection of input documents, LDA refers to a single document as a mixture of different and latent topics, and each topic is represented as a probability distribution over words. The words with highest probabilities in each topic usually give an idea about that topic [21]. Given a corpus D a collection of M documents denoted by D = (W1,W2 … Wm), and each document is represented as a sequence of N words denoted by W = (w_1,w2, …, w_n), where w_n is the nth word in the sequence. The generative process for each document W in the corpus D is given as follows [26]:

1. select $N \sim \text{Poisson}(\xi)$
2. select $\theta \sim \text{Dir}(\alpha)$
3. For each word w_n in a document:

3.1. select a topic $z_n \sim \text{Multinomial}(\theta)$

3.2. select a word w_n from $p(w_n|z_n,\beta)$, a multinomial probability conditioned on the topic z_n

A k-dimensional Dirichlet random variable θ can take values in the $(k-1)$ simplex (a k-vector θ lies in the $(k-1)$ simplex if $\theta_i \geq 0, \sum_{i=1}^{k}\theta_i = 1$), and has the following probability density on this simplex:

$$p(\theta|a)=\frac{\Gamma\left(\sum_{i=1}^{k}a_i\right)}{\prod_{i=1}^{k}\Gamma(a_i)}\theta_1^{a_1-1}\ldots\theta_1^{a_k-1} \tag{4}$$

where the parameter α is a k-vector with components $\alpha_i > 0$, and where $\Gamma(x)$ is the Gamma function. Given α and β parameters, the joint distribution of a topic mixture θ, a set of N topics $\mathbf{z}$, and a set of N words $\mathbf{w}$ is given by:

$$p(\theta,\mathbf{z},\mathbf{w}|\alpha,\beta)=p(\theta|\alpha)\prod_{n=1}^{N}p(z_n|\theta)p(w_n|z_n,\beta) \tag{5}$$

where $p(z_n|\theta)$ is simply θ_i for the unique i such that $z_n^i = 1$. Integrating over θ and summing over z, we obtain the marginal distribution of a document:

$$p(\mathbf{w}|\alpha,\beta)=\int p(\theta|\alpha)\left(\prod_{n=1}^{N}\sum_{z_n}p(z_n|\theta)p(w_n|z_n,\beta)\right)d\theta \tag{6}$$

Finally, taking the product of the marginal probabilities of single documents, the probability of a corpus is given by:

$$p(D|\alpha,\beta)=\prod_{d=1}^{M}\int p(\theta_d|\alpha)\left(\prod_{n=1}^{N_d}\sum_{z_{dn}}p(z_{dn}|\theta_d)p(w_{dn}|z_{dn},\beta)\right)d\theta_d \tag{7}$$

Fig. 7 illustrates the graphical representation of LDA divided into three main levels. The parameters α and β are corpus-level parameters which are set in the process of generating a corpus. The variables θ are document-level variables parametrized once per document. Finally, the variables z and w are word-level variables parametrized once for each word of each document.

4.4. Lda2Vec

Lda2Vec was introduced by Christopher Moody[73] in 2016 as a hybrid topic model that integrates Word2Vec representation with LDA. This approach is designed to learn dense word vectors jointly with Dirichlet-distributed latent document-level mixtures of topic vectors.

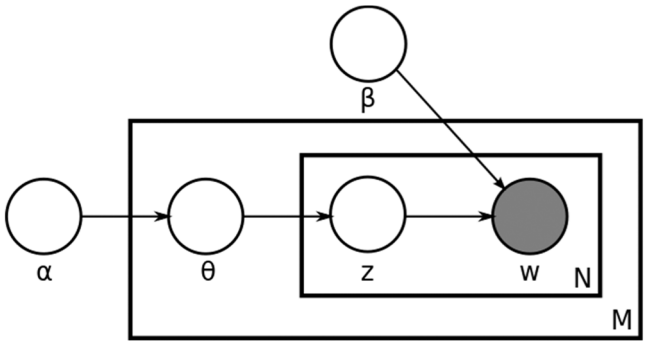


Fig. 7. Graphical model representation of LDA.

The formulation of word embeddings and LDA produces sparse and interpretable mixtures of documents using a non-negative simplex constraint. The power of Lda2Vec over other topic models lies in learning topic representations and document representations at the same time. The model begins by generating word vectors on top of a Skip-Gram architecture (in which we try to predict the surrounding words based on the central word), and then, context vectors are created from the element wise sum of the word vectors and the document vectors. The document vector is, in turn, created from a combination of two main components:

- Document weight vector that represents the "weights" of each topic in the document.
- Topic matrix that represents each topic and its corresponding vector embedding.

Classic topic models do not benefit from recent advances in distributed word representations like Word2Vec which has the ability to capture semantic relationships between words of a document. The examination of word co-occurrences has shown to be an effective method. The experiment results after training and testing the model on several datasets like 20Newsgroups and Hacker News comments corpus showed high mean topic coherence. In addition, Lda2Vec showed high performance in discovering latent topics within the Hacker News comments corpus and effectively learns linear relationships between words [73].

4.5. BERTopic

BERTopic[74] is a topic modeling algorithm that leverages the power of BERT (Bidirectional Encoder Representations from Transformers), a state-of-the-art language model, to extract topics from textual data. It combines the strengths of BERT's contextual embeddings and a technique called Hierarchical Dirichlet Process (HDP) for topic modeling. BERTopic has gained attention in the research community due to its ability to generate coherent and interpretable topics from unstructured text [75]. The algorithm works by first transforming documents into numerical representations using BERT's pre-trained embeddings. These embeddings capture the contextual meaning of words within a document. BERTopic applies HDP technique, which is a non-parametric Bayesian approach, to discover the underlying topics in the collection. HDP allows the model to automatically determine the number of topics without needing prior specification.

One notable advantage of BERTopic is its capability to handle large-scale datasets efficiently. The topic model utilizes the uniform manifold approximation and projection (UMAP) dimensionality reduction technique to reduce high-dimensional BERT embeddings into a lower-dimensional space. This reduces the computational complexity and speeds up the topic modeling process. BERTopic has been applied in various domains such as social media analysis, healthcare, and customer reviews. It has shown promising results in extracting meaningful topics from diverse text sources [76]. Researchers have also explored extensions and variations of BERTopic by incorporating domain-specific information and evaluating its performance against other topic modeling methods.

4.6. Non-negative matrix factorization (NMF)

NMF is a matrix decomposition method used in topic modeling, and also applied for dimensionality reduction. It decomposes a document-term matrix into two lower-dimensional non-negative matrices, which represent the document-topic and topic-term distributions [77]. This technique ensures that the results are interpretable because of the non-negative constraint. Moreover, this method is simple and flexible, but it can suffer from the initialization sensitivity of the parameters and the scalability on large datasets[78]. NMF based topic models do not rely on model or data-assumptions, making them seen as difficult optimization problems. In their work, the authors proposed a deep NMF (DNMF) topic modeling framework to alleviate this issue [79]. The proposed model uses an unsupervised deep learning approach to extract latent hierarchical structures from documents and utilizes them to constrain topic-document distributions, which enhances topic discovery and reduces computational complexity.

NMF is a specific technique often used to perform Latent Factor Analysis (LFA) for text mining tasks like topic modeling, to uncover latent topics in documents[80]. LFA is a statistical method that identifies hidden or latent factors influencing the relationships among observed variables. Peladeau Normand and Davoodi Elnaz [81] presented a comparison of LDA and Factor Analysis (FA) for topic extraction. Using a new evaluation method, the authors found that FA may produce more coherent topics than LDA. LFA can be beneficial when dealing with large datasets such as textual ones where the relationships between variables are too complex to capture without uncovering the hidden factors [82]. Recently, several state-of-the-art latent factor analysis models [83–85] have emerged to address the challenges of high-dimensional and sparse data. For instance, Wu et al. [83] proposed prediction-sampling-based multilayer-structured latent factor (PMLF) analysis which offers a robust approach to handling sparse and high-dimensional datasets by utilizing multilayer structures and prediction-sampling techniques. These models indicate the growing diversity and sophistication of latent factor analysis techniques, which are not only useful for topic modeling but also play a crucial role in its application. For example, in recommendation systems, the collaborative filtering approach uses latent factors to predict missing user preferences, based on existing ratings or interactions [82,86].

4.7. Correlated topic model (CTM)

CTM can be considered as an extension of traditional models like LDA that allows topics to exhibit correlations [87,88]. Unlike LDA, which supposes that topics are independent, this technique tries to capture the dependencies and relations between topics. This is why it is effective when documents contain related topics. Consequently, it enables a richer representation of topics structure, making it very useful for documents where topics are interlinked and may influence each other.

4.8. Top2Vec

Top2Vec [89] is a modern topic modeling technique as all the most modern approaches use word embeddings to find topics. Unlike other methods, Top2vec leverages word embeddings to identify the themes directly from embeddings, finding topics that are semantically more meaningful. This technique is effective for large datasets and provides better results for topic coherence and document similarity [90]. Top2vec is able to identify hierarchical structures in topics, making it easier to find the main topics and their subtopics. Moreover, Top2Vec can effectively handle the problems of synonymy and polysemy. However, it may be computationally intensive.

4.9. Other topic models

Topic modeling has received a growing interest from the community of NLP researchers in past few years. Many studies have been done to advance and improve topic models efficiency in discovering themes from a given collection of documents. In 2014, Yin and Wang proposed [91] a topic model called Gibbs sampling Dirichlet multinomial mixture (GSDMM) for short texts based on Dirichlet multinomial mixture (DMM) model [92]. The model was built on top of the original DMM using Gibbs sampling algorithm to improve its scalability. In another work, Li et al. [93] designed a common semantics topic model (CSTM) for filtering noise from topics in social media datasets. The authors added a mixture of unigrams to the proposed model known as common topics. These

common topics are designed to capture common words that occur in all topics. Each document is generated from a function topic and a set of common topics. Bunk and Krestel [94] proposed a word embedding LDA (WELDA) which is a combination of word embeddings and LDA. The authors used a pre-trained embedding model as a vector space to build an altered version of LDA's generative algorithm for topic extraction. In 2020, Thompson and Mimno [95] proposed a new topic model using BERT[64] to extract topics from documents. They used k-means algorithm to group observed tokens in the collection based on their contextual vectors created from BERT. The integration of knowledge graphs (KGs) with topic models is an effective approach to enhance the contextual and semantic relevance of topics [96]. It makes the models capture deeper connections among concepts instead of using only word co-occurrence which lead to the loss of significant semantic information.

In the work of Ennajari et al. [97], the authors proposed a Bayesian embedded spherical topic model (ESTM) that combines knowledge graph and word embeddings to enhance topic interpretability and capture discriminative text representations. Liu et al. [98] proposed K-BERT that utilizes the KGs to perform transfer learning on the pretrained BERT model. According to Min et al. [99], this model can cause confusion in the training. Hence, the authors introduced TK-BERT model that uses KGs topic model to conclude the topic in the input sentence. This model outperformed K-BERT model because it does not add unnecessary knowledge to KG. Although the use of KGs in topic modeling improved the interpretability of topics, it may seem expensive in terms of computation complexity in addition to its limitation to domain specification. In another recent work, Gao et al.[100] introduced a modern neural topic model (NTM) that enhances topic interpretability through contrastive topic-wise learning process, fostering coherence and diversity. Another topic model, known as TopicGPT, was recently presented by[101] that leverages the powerful capabilities of large language models to discover latent topics in a text corpus. This model uses a prompt-based strategy to generate topics that closely align with human judgements, achieving a harmonic mean purity score of 0.74 when compared to human-annotated categorization. Table 2 shows the advantages and disadvantages of five popular topic models.

Table 2
Advantages and disadvantages of some topic models.

Model	Advantages	Disadvantages
LSA	• Reduces dimensionality of term document matrix effectively • Captures synonymity and semantic similarity between words of the corpus • Mitigates the effect of noisy texts and improves it through text representation	• Loss of interpretability which makes generated topics less coherent and understandable • Disregards order and context in BOW model which leads to not retaining semantic relationships between words • Polysemy is not handled efficiently while representing words with a single vector
NMF	• Enforces non-negativity of in the factorization process which benefits the interpretability of topics • Produces a sparse representation in which each document is seen as set of topics and each topic is represented by a subset of words • Term document matrix can be weighted by TF-IDF values • Easy to implement	• Different setups may result in different factorization meaning different results. • Sensitive to missing data which requires performing preprocessing tasks for incomplete data. • Number of topics must be predefined • Often produces topics with a low coherence score
LDA	• Provides a probabilistic generative framework for capturing the structure of topics in a corpus • Scalable and effectively tolerant to large collections • Captures the co-occurrences between terms of the corpus which enables identifying relationships between them • Generates multinomial distributions over topics • Does not require prior domain knowledge	• Requires setting the number of topics in advance • Low performance in collections of short documents • Sensitive to some preprocessing tasks such as stop words removal and stemming which impacts topic modeling • Hyperparameter tuning is necessary • Produces overlapping topics
CTM	• Models topic correlations and identifies relationships between topics • A single document can be related with multiple topics • Handles large scale collections effectively	• Requires high computational compared to baseline models • Difficulty in capturing interpretability of correlations between topics • Choice of hyperparameters can affect the results • Does not capture such complex relationships between topics
BERTopic	• Captures the contextual of terms within documents through BERT-based embeddings • Prior number of topics is not defined • Provides a tool for visualizing topics distribution • Deals with multilingual collections • Supports various types of topic modeling such dynamic, guided, and class based • Number of topics is automatically identified	• computationally efficient especially when dealing with large corpora • Requires high computation resources of memory and processing • The quality of results relies on the pretrained model • Generates high amounts of outliers • Missing of objective evaluation metrics

5. Evaluation metrics

The evaluation of topic models still remains the most challenging task due to the unsupervised nature of topic modeling. Nevertheless, it is important to assess the performance of such topic model and have the ability to compare one against the other given some criteria. Besides coherence, perplexity, and diversity, many quantitative metrics and qualitative measures are used to evaluate topic models. To assess the performance of topic modeling techniques, researchers have been working on datasets like 20Newsgroups and Neural Information Processing Systems (NIPS). This section covers the most commonly used evaluation metrics for topic modeling. These metrics can be divided into two types: intrinsic evaluation, which assesses the model's internal characteristics and extrinsic evaluation which measures how well the topics align with real-world data or human interpretation.

5.1. Coherence

Topic coherence measures the degree of semantic similarity between high scoring words that constructs a generated topic. Measuring coherence helps to differentiate between topics which are semantically interpretable and topics coming out of a statistical inference. Coherence methods attempt to grasp the usefulness of these individual topics and their words. Many evaluation metrics are used in early works to leverage coherence of topics in a collection of documents. Coherence precision calculates the fraction of the top words from each topic that belong into the best fitting ground truth topic. Another coherence metric is defined as pointwise mutual information (PMI) which leverages the similarity of words in each topic based on their relative co-frequencies with each other. The PMI between a pair of words (w_i, w_j) is computed as follows:

$$PMI(w_i, w_j) = \log \frac{p(w_i, w_j)}{p(w_i)p(w_j)} \tag{8}$$

PMI is calculated by measuring the similarity between all combinations of pair words in each generated topic. Other coherence metrics have been used to evaluate topic models like C_V [102] which combines both cosine similarity and normalized PMI. This metric tries to capture the mutual information and vector similarity as well as the proximity between words.

5.2. Diversity

Topic diversity measures the degree of semantic diversity among generated topics by a topic model. Many different metrics are presented to compute topic diversity. In [103], the authors used the average pairwise Jaccard distance to leverage topic diversity between extracted topics from small corpora. Dieng et al. [104] used the proportion of unique words (PUW) to measure semantic diversity among generated topics. They define the percentage of unique words in the top 25 words of all extracted topics. Diverse topics are obtained with value of PUW close to 1 and redundant topic are generated with a value close to 0. Bianchi et al. [105] used inverted rank biased overlap (IRBO) to evaluate diversity among topics in a single model. This metric compares the top 10 lists of words in every two topics having different terms. Topics are considered completely diverse when IRBO reaches 1. Topic diversity is directly affected by the number of pre-defined topics because a large number of topics leads to identical topics with overlapping terms and small number may cause poor interpretability.

5.3. Perplexity

Many topic models are still facing the challenge of assuming the optimal number of topics to be generated to obtain good results. Perplexity is an optimization technique used to determine the optimal number of topics in advance. One way to measure perplexity is to compute the log likelihood between text documents and generated topics. A good topic model scores high log likelihood meaning low perplexity score. Another way to measure perplexity is by calculating the rate of perplexity change (RPC) over many runs or iterations and the resulting value is used as a hyperparameter to determine the optimal number of topics. The authors in [106] have divided data into a random number of m samples $\{S_1, S_2,\ldots, S_m\}$ and trained topic models with different topic numbers (from t = 1 to r) using m-fold cross validation. The means of perplexities obtained from the m samples for each topic number t are denoted by $P_1, P_2,\ldots,P_r$. Finally, the positive value of RPC for a topic number t_i where $0 < =i < =r$ is given by the formula below:

$$RPC(i) = \left|\frac{P_i - P_{i-1}}{t_i - t_{i-1}}\right| \tag{9}$$

Perplexity is often prone to instability and its values may remarkably differ between iterations over the same sample dataset. Therefore, perplexity may not always be the best choice to determine the optimal number of topics for a topic model. However, RPC method has demonstrated a significant performance and still requires an iterative process of generating topic models in order to optimize the number of topics.

6. Comparison of topic models

In this section, we conducted a set of experiments to compare some topic models, namely, LSA, NMF, LDA, ProdLDA, BERTopic, and CTM over three different datasets. The selection of these models is purely experimental and does not follow any predefined criteria. That said, we aimed to select models of different categories, including two algebraic models, two probabilistic models, one neural model, and one hybrid model to evaluate their performance across all datasets. In Table 3, we show the characteristics of the three datasets.

Table 3
The characteristics of the three datasets.

Dataset	Language	Content Type	Number of Categories	Size
20Newsgroups	English	Posts	20	20000
BBC News	English	News articles	5	2225
Hespress News	Arabic	Articles	5	100000

We used python packages and frameworks such as Gensim,[1] OCTIS[2] and Pytorch[3] to train, test and optimize topic models. For the preprocessing tasks like tokenization, stemming, stop words removal, etc., we used natural language toolkit (NLTK) python package.[4] The experiments were carried out by controlling two main factors: the number of topics and the number of iterations (or runs). For evaluation, we used topic coherence (TC) to assess the semantic similarity of words within each extracted topic, highlighting its importance for topic interpretability. On the other hand, we used topic diversity (TD) to measure the degree of distinctiveness across all extracted topics ensuring that the model avoids generating redundant or overlapping topics. For the 20Newsgroups dataset, we achieved high scores of TC and TD across 5 runs with 50 topics. Similarly, we observed high TD and TC for BBC News over 10 runs with 5 topics while high scores are recorded for Hespress News over 4 runs and 5 topics. Table 4 presents our experimental results while Table 5 shows the best performing model for each dataset.

In Table 4, we notice that BERTopic has a high coherence score while ProdLDA has a high diversity score in the 20Newsgroups dataset. This means that for this dataset, LDA generated coherent and interpretable topics while ProdLDA excelled in extracting diverse and non-redundant topics. In BBC News dataset, BERTopic outperformed other models in coherence and CTM has high diversity degree. In the Arabic Hespress dataset, NMF has a high coherence and diversity scores.

By examining these results, we observe that the performance of some topic models varies when the training dataset is altered. This variation in results can be referred to many factors such as dataset domain, length of texts, or some characteristics of the model itself. As for now, researchers have no predefined guiding criteria on how to select a topic model for a specific application. However, there are some proposed aspects to consider when choosing a topic model that specifies the relevance of the evaluation criteria [24]. One of these aspects checks whether if the generated topics will be fed to humans or machines. Another aspect considers corpus coverage and whether if the model's output covers the existing diversity in the input collection or not. In some applications of topic modeling, another challenge occurs in the moment of choosing the appropriate evaluation metric to asses a topic model. For instance, researchers in social studies might consider coherence over diversity and stability to assess an algorithm. In another application such as text classification and text summarization, it is preferred to use diversity and efficiency over coherence.

As mentioned before, traditional techniques may not work well for short texts problems. In this case, several solutions have been proposed to address this issue. Yan et al.[50] proposed Biterm Topic Model (BTM), a model that learns topics by directly modeling the generation of word co-occurrence patterns in the entire corpus. Similarly, He et al. [52]

Table 4
comparison of topic models over the three datasets.

Dataset	20Newsgroups		BBC News		Hespress	
Model	TC	TD	TC	TD	TC	TD
LSA	0.0128	0.4810	0.0123	0.577	−0.019	0.423
NMF	0.0821	0.6945	0.0124	0.549	0.158	0.951
LDA	0.0575	0.7105	0.0142	0.577	−0.021	0.532
CTM	0.0547	0.820	0.0494	0.839	0.023	0.872
ProdLDA	0.0552	0.8785	0.0131	0.814	0.0115	0.682
BERTopic	0.096	0.851	0.167	0.794	0.093	0.844

[1] https://radimrehurek.com/gensim
[2] https://github.com/MIND-Lab/OCTIS
[3] https://pytorch.org/
[4] https://www.nltk.org/

Table 5
Best and worst topic models across three datasets.

Dataset	Best models		Worst models	
	Coherence	Diversity	Coherence	Diversity
20Newsgroups	BERTopic, NMF	ProdLDA, BERTopic	LSA	LSA
BBC News	BERTopic	CTM, ProdLDA	LSA, NMF	LSA, NMF
Hespress	NMF	NMF	LDA, LSA	LSA

proposed FastBTM, an amelioration of the previous model that is faster, more effective and work well for both short text and long document datasets. In the work of Wu et Li [51], the authors introduced a Multi-term Topic Model (MTM) for short text topic modeling. This approach extracts variable-length and more correlative word patterns (multiterms instead of Biterm) from the whole corpus.

7. Applications of topic modeling

Topic modeling is a powerful technique in NLP that helps to uncover latent patterns or themes within large text corpora. It is primarily used to automatically identify topics in a collection of documents without the need for manual annotation or pre-defined labels. This review stands out from other reviews as it discusses various applications of topic modeling across different domains.

7.1. Information retrieval

Information retrieval (IR) is a set of techniques that are designed to retrieve information from unstructured or semi-structured data sources grouped as a collection of documents. Many classic information retrieval systems (IRS) or search engines use term weighting techniques like TF-IDF to only measure the lexical query-document matching. However, these methods don't take in consideration the contextuality and the semantic aspect of words within documents. Recent works have integrated topic modeling techniques in the information retrieval process to address some limitations of classic IRSs and improve the information search. Topic models measure the query-document matching in a generated feature latent space where terms are semantically related.

In his paper [107], Thomas Hofmann proposed a probabilistic version of latent semantic analysis model (PLSA) to overcome the problems in classic latent indexing IR-based models. He combined cosine similarity function with a straightforward representation based on term frequencies. Wei and Croft introduced in [108] a term-weighting scheme computed in the latent space of document topics based on LDA. The evaluation of topic modeling is based on the query likelihood model[109] where each document is weighted by the likelihood of its topic model to generate the query. The LDA-based model computes the query likelihood from each topic in the document using maximum likelihood estimators directly on a collection of documents in the classic IRSs. Thus, two documents with lexical matching with the query are ranked differently based on document's topics distribution.

7.2. Text classification and clustering

Unstructured textual data can be a rich source of information, and yet the task of extracting this valuable information remains hard and time-consuming. Topic models are designed to detect the main themes (i.e., topics) within a collection of documents, then assign these topics to documents. Text classification is another important NLP task that aims to assign a given text to a specific category or class (i.e.; sport, economy, art, tech, etc.). Topic modeling techniques are effectively used to classify documents by enhancing the topic extraction task. In a topic modeling approach, every document in the corpus is represented as a mixture of topics, and, in turn, each topic is considered as a set of words that can tell what the topic is about, and from then, knowing what the documents are talking about. Additionally, topic models are also used in text (or document) clustering to group documents with semantic similarities [110]. Topic modeling projects input texts in an embedding vector space to facilitate clustering documents into similar groups called clusters. Topic models try to extract the featured topics in a collection, and assign each topic to a particular cluster. BERTopic[74] works following the same process in transforming a corpus of texts into different clusters based on their proximity in the embedding space, where each document cluster represents a specific topic. For each cluster, BERTopic identifies the most representative words of each extracted topic. In a recent work [111], Sánchez-Franco et al. applied BERTopic to cluster Scopus paper abstracts on quality management (1980–2020), identifying 48 topic-clusters and 13 meta-topics. The authors also showcased the evolution of these topics over time while emphasizing on the growing interest in quality management literature.

In[112], Chen et al. proposed an improved short text classification method that combines LDA and K-Nearest Neighbor algorithm (KNN) applied on a short dataset containing posts from Sina News website. Experiment results showed that their proposed hybrid model performs well compared to other models. Burkhardt and Kramer [113] proposed new model based on LDA to perform multi-label text classification. Their proposed method uses a greedy neural-network-based strategy to train the model one layer at a time. Experiments revealed a good performance in modeling dependencies online for large-scale multi-label datasets with many labels and improvements were made over the baseline methods.

7.3. Recommendation systems

Recommendation systems (or recommender systems) are a type of information filtering systems that suggests items that are most pertinent to a particular user given his history of interactions with other related items. The suggestions generated by recommender systems are based on various decision-making processes, such as what product to purchase, what music to listen to, or what movies to watch, etc. Recommender systems help individual users to make relevant choice of an item from a number of items that a particular service may offer. There are many types of recommendation systems, but two of them are the most common: content-based and collaborative [114]. The content-based filtering suggests items based on their characteristics and the second recommends them based on the history of user's behavior as well as similar decisions made by other users. Topic modeling techniques are a reliable solution to many recommender systems to improve the predictivity of suggested items. Rajendran and Sundarraj [115] applied LDA-topic models on Wikipedia data to generate topics, and use them later on the user's navigating history. These topic models are then embedded in a collaborative filtering recommender system to extract user's preferences. Experiments in two different domains, movies and restaurants, showed an enhancement in gathering user ratings and their browsing history.

7.4. Question answering

Topic modeling can serve as a powerful tool in question answering (QA) systems, due to its ability to identify the underlying topics in questions. This can make QA systems more accurate in retrieving contextually relevant information, hence enabling responses that are more tailored to user inquiries. As an example, categorizing questions into a specific topic streamlines retrieval in large datasets and improves response precision. In a work of Prebha and Sardana [116], the authors used LDA to identify the latent topics that accurately represent the content of documents sourced from Community Question Answering (CQA) websites such as Quora, Stack Overflow, and Yahoo. Similarly, Chen el al. [117] used a hybrid approach that combines classic Query-Likelihood language model, translation-based language model and a proposed semantic graph-based topic model. The findings showed

that the proposed approach significantly outperforms existing question retrieval techniques. As a last point of this subsection, we can see that topic modeling in question answering intersects with recommender systems, particularly in personalized information retrieval. Topic modeling can make recommendation systems more effective if they are aligned by utilizing user inquiries with content themes. This intersection opens pathways for cross-disciplinary applications in personalized assistance and contextual recommendations.

7.5. Bioinformatics

Bioinformatics is another promising application field for topic models with the increasing volumes of biological data like microarray datasets [118]. The task of extracting hidden knowledge in these datasets yet remains challenging. However, Topic modeling can be useful to model biological objects as hidden topics that can reflect the underlying biological meaning more comprehensively.

In their work, Sinkkonen et al. [119] developed an infinite topic-based model to find functional gene modules (i.e.; topics) applied on gene expression data. The model focuses on the relation among documents to be suitable for clustering of other relational data. In another study, Chen et al. [120] analyzed the genome-level structure of DNA sequences using LDA. They represented the DNA sequences by N-mer frequencies in the first step, and in the second step, they considered genome sequences as documents, and the N-mers are represented as words, and finally, they discovered the genome-level statistical patterns as topics.

7.6. Text summarization

Text summarization is a text mining technique that aims to reduce the number of sentences in a given document. There are two common types of strategies to summarize a text: extractive and abstractive. The extractive method identifies the most important sentences within a text and resulting summary which is a subset of the original text. The abstractive approach uses advanced NLP techniques like word embeddings and attention-based transformers to capture the semantics within a text and generate a relevant summary. Topic modeling techniques (LSA, LDA, etc.) are widely used to perform an extractive text summarization task. Wu et al. [121]used topic models to extract the candidate sentences with topic words from a preprocessed text of a novel, and an importance evaluation function is designed to construct an initial summary. This initial summary is then passed to a smoothing function to overcome the semantic confusion due to ambiguity or synonymity of words, so as to optimize the readability of the summary. Roul et al. [122] proposed a topic modeling methodology based on LDA to generate extractive summary from a corpus of documents. Their approach is designed to produce a summary that covers the most important in a corpus. The important sentences are identified from each topic using word and sentence level features.

7.7. Opinion mining and social network analysis

Opinion mining (also called sentiment analysis) has become a popular research area due the huge volumes of user-generated data in web platforms such as blogs, forums, social media outlets. etc. Sentiment analysis is defined as a set of NLP techniques that aim to classify a subjective text into positive, neural, or negative. Several works have been proposed [123–125] to handle sentiment analysis problems from a topic modeling perspective, aiming to improve sentiment classification. Researchers can develop sentiment-aware topic models that incorporate sentiment information into the topic modeling process. This involves extending existing topic models such LDA or LSA to incorporate sentiment indicators or sentiment lexicons and uncover topics or themes associated with specific sentiments in the text.

Topic models can be used to identify the latent aspects in a corpus of reviews or customer feedback [126]. By combining topic modeling with sentiment analysis techniques, sentiment polarity is linked with specific aspects to perform a fine-grained analysis. In some cases, topic models can be implemented to extract topics from a corpus of documents, and then sentiment analysis techniques are applied to assign sentiment polarities regarding these topics [127]. By applying sentiment analysis techniques to the text and clustering documents based on the expressed sentiment, we can cluster groups of documents that share similar sentiments. These sentiment-driven clusters can then be treated as topics or themes within the dataset.

8. Challenges of topic modeling

Although topic modeling has achieved promising results across different domains including information retrieval, social network analysis, and bioinformatics, academic research is still ongoing to overcome some challenges known in topic modeling. As the demand for more sophisticated, scalable, and accurate topic models grows, understanding and addressing these challenges remains a critical area of research. In this section, we explore the key challenges faced in topic modeling, examining both technical limitations and practical considerations that hinder its broader application and effectiveness. One of these challenges arises during the text preprocessing phase, which often involves dealing with a lot of noise such as spelling errors, colloquial language, abbreviations and hashtags, etc.[128]. Preprocessing texts involves cleaning, tokenization, stemming, part of speech tagging and other NLP tasks that can directly affect the quality of generated topics, topic coherence, topic diversity and stability. Choosing the right number of topics in topic modeling remains one of the primary challenges for researchers and practitioners alike, because it can significantly impact the performance of downstream tasks [129]. This challenge arises from the inherent ambiguity and variability in textual data. Since there is no absolute correct number of topics in a dataset which can vary widely based on the nature of the dataset, it depends on the researcher's judgment [130]. Thus, the selected number should balance between specificity and generality. Moreover, the evaluation metrics used for topic modeling do not align with human interpretation of topic quality. In this situation, sometimes there is a need for an often lengthy and repeated process of trial and adjustment to find the right topic number that can be very time consuming.

Another topic modeling challenge is faced when dealing with user generated contents (UGC) in social media such as tweets and Facebook posts. These texts are relatively short and their representation may lead to generate sparse and large dimensional vectors which makes it difficult for topic models to capture semantic relationships between words [131]. Short texts are also characterized by the lack of contextual information which affects topics interpretability [132]. Topic models often generate different topic distributions over multiple iterations on the same dataset with the same parameters which highly affects model stability, especially when handling large collections. Classical topic models such as NMF and LDA still suffer from instability when these models are retrained on the same samples of data [133,134] and on short text collections. In most cases, achieving interpretable and explainable topics is one of the critical challenges posed to topic modeling [135]. Interpretability refers to the human ability to understand generated topics by the model and obtaining interpretability means to ensure generating coherent, accurate, and meaningful topics. In addition to the above, another issue arises in the choice of the appropriate model due to the existence of several parameters to consider. In many times, researchers rely on different collections and different evaluation metrics to test these algorithms [136]. As a result, the comparison between these techniques can be inconsistent and inaccurate because of lacking evidence-based criteria to choose the right topic model [137]. Moreover, the choice of the right metric to evaluate topic models is also challenging due to the particularity of the application domain. Some metrics like recall and PMI coherence depend on statistical accuracy of the clustering results. These

metrics are not always the optimal assessment to evaluate the quality and the interpretability of a topic model in such domains like social sciences [138].

One of the nuanced challenges in topic modeling is the issue of overlapping topics or probabilistic topic modeling. It refers to the case where documents or sets of words may be used for multiple themes simultaneously. In natural language, documents usually are composed of cross-cutting themes. Traditional methods such as LDA may struggle capturing these themes as they assign documents a dominant topic [139]. The presence of overlapping topics makes the model fail to represent the real nature of the context, adding more difficulties in interpretation. Several tasks that rely on accurate topic modeling are highly affected by overlapping topics. For instance, in recommendation systems, documents may force into rigid categories that do not reflect their true thematic nature. This will lead to poor user experiences. Although, newer approaches such as Non-negative Matrix Factorization (NMF) [77] or neural-based models (e.g., BERTopic) offer improved ways to capture overlapping topics, they came with their own challenges such computational complexity. Therefore, a sophisticated technique for addressing the overlap is still needed. As we are dealing with the natural language, synonyms and polysemy are always the big challenges [49], especially in topic modeling tasks. Handling this problem is essential to create more accurate and coherent topics. However, traditional models have limitations in understanding word relationships and contextual meanings.

Lastly, when dealing with dynamic datasets such as news articles, social media posts, or scientific publications, tracking and understanding evolution of topics over time is significant [17,140]. Creating a model for alike task should take into consideration several other challenges including capturing temporal shifts and the problem of data segmentation. Dynamic Topic Models [16] and other temporal extension used to address this challenge. However, these models face several other challenges, particularly the implementation and the computational complexity.

9. Conclusion and research directions

The large amounts of UGCs in the internet arises the need to process large collections of text in real time more indispensable. Topic modeling has been an interesting research area in recent years. In this paper, we presented an overview about topic modeling techniques categorized into algebraic, probabilistic and neural. We also introduced the most common models of text representations such as BOW and word embeddings. The most common applications of topic models across different areas are summarized and discussed. The authors showed how topic models are very useful to perform many NLP tasks such as topic extraction, topic classification, text summarization, sentiment analysis, etc. We introduced some evaluation metrics which are utilized to assess the quality of topic models and we compared the performance of some algorithms across three different datasets. Several challenges of topic modeling are discussed to show the current limitations related to the particularity of some algorithms, evaluation metrics, and preprocessing. In order to overcome these limitations, researchers in the field are still working to design coherent, interpretable, and stable models and to improve the evaluation criteria for better results.

In addition to the challenges above, many opportunities exist in topic modeling. According to [136], The research directions in this field can be broadly grouped into three main categories: developing better models, creating automatic evaluation methods, and enhancing data discovery through multi-disciplinary applications. In the context of the first group, research efforts focus on improving the stability and the coherence of topic models. This can be done through the regularization of models and fine-tuning hyperparameters to optimize performance. Creating models that adapt effectively to various types of data is also considered in this category without ignoring the enhancement of computational efficiency allowing faster processing and scaling of models, making TM more practical for large datasets. While human evaluation usually considered the most reliable form of assessment due to its qualitative insight, it may be time-consuming and very expensive. Thus, evaluating topic models effectively is an ongoing research area. Studies in the field use varying evaluation processes and metrics which makes it difficult to compare models consistently. This explains the need for a unified and widely accepted evaluation framework for topic models. Because of its versatility, the third areas of research focus on expanding the use of topic modeling in diverse research areas to open new opportunities for innovation and application. Other disciplines such as political science, psychology and sociology employ topic modeling to extract meaningful themes and patterns from the available data to generate hypotheses.

The research in this filed is not limited only to these three areas. Research direction goes also to develop more advanced techniques to enhance interpretability of generated topics by considering domain specific knowledge. Other future research can develop multimodal topic models by extracting information from different modalities (i.e., text, image, audio) for a good understanding of document content. In another level, combining topic models with semi-supervised and active learning techniques helps gaining insights from labeled texts and transfer this knowledge to improve some tasks like document classification. To extend our review, future work is going to compare deep learning (DL) and transformer-based techniques in topic modeling against conventional approaches such as probabilistic models (LDA, PLSA, CTM). This comparative study will provide both academics and practitioners with a comprehensive and holistic overview for DL-based topic models in the text mining.

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CRediT authorship contribution statement

Mustapha Hankar: Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Abderrahim Beni-Hssane:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Conceptualization. **Mohammed Kasri:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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